Local Rules for Protein Folding on Triangular Lattice in the 2D HP Model

(Extended Abstract)

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Abstract

Agarwala et al. *proposed two main folding rules, namely,* backbone folding *and* arrow folding*, on triangular lattice in 2D HP model [1]. However, these folding rules are not suitable for sequences with sparse H's. In this paper, we propose two folding rules, which can support the weakness of their folding rules. The first rule is called* starlike folding *which is better than others for the sequences in the pattern of (HP$^+$)$^+$. The other rule is called* compact starlike folding *which is suitable for the sequences in the combined patterns of (HP$^+$)$^+$ and H^n. Together with backbone folding and arrow folding, we propose a general strategy of protein folding which suggests a suitable usage for all possible sequences.*

1 Introduction

A protein is a sequence of amino acids linked together. There are twenty different amino acids used in the formation of proteins. The length of a protein sequence ranges from 50 residues to about 2000. Understanding the behavior and properties of proteins is one of the most fascinating challenges in molecular biology.

After its discovery, many scientists attempt to answer two questions: How should people predict the structure of a protein given its sequence? Is it possible to find a protein sequence that folds into a desired structure? Both questions, known as *direct* and *inverse foldings*, remain unanswered. Much effort (about thirty years worth) has been expended trying to understand how proteins fold up in nature. Understanding structures allows biologists to make an educated guess about protein functions. It would also enhance our ability for drugs design by seeing the three dimensional molecular details of the proteins in our body with which almost all kinds of drugs interact.

In this paper, we study the Hydrophobic-hydrophilic (Polar) model introduced by Dill [9]. It is one of the most studied models in this domain [3, 4, 6, 10, 15, 17]. The HP model abstracts the problem by dividing the 20 amino acids into two classes: hydrophobic and hydrophilic residues. We can classify all amino acids in a protein as hydrophobic or hydrophilic [7, 20].

In this model, we only consider a protein sequence from {H, P}, where H represents hydrophobic residues, and P represents polar residues. For convenience, we use regular expressions to describe these sequences. For example, H^n indicates the sequence with n H's in sequence, H^3(P$^+$H)$^+$ indicates 3 H's following at least one subsequence with (P$^+$H) pattern where (P^+H) means that the subsequence contains at least one P and ending on an H.

The Hydrophobic-Polar model is a free energy model. The energy function for this system favors interactions between H-H monomer types and is indifferent to all other interactions (P-P and H-P) . This is known to highlight the importance of the hydrophobic effect in protein folding. A nonadjacent H-H bond is called a *contact.*

The HP model is also a lattice model because those HP sequences are placed in a lattice. The lattice nodes are responsible for keeping the nodes of sequence at the fixed lattice node position. The characters of lattice model are as follows.

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- Monomers are represented using uniform size.
- Bond length is uniform.
- Positions are restricted to a regular lattice.
- Every conformation is a self-avoiding walk in 2D or 3D.

There are several types of lattice models. For example, 2D square, 2D triangular, 2D hexagonal, 3D cubic, 3D triangular and 3D diamond, etc [14]. The most used lattice is square lattice. However, square lattice has parity constraint, i.e., the sequence $(\mathrm{PH})^n$, has 0 contact, but has $2n-3$ contacts on the triangular lattice. This sequence is the worst case on the square lattice but is the best case on the triangular lattice. Thus, the triangular lattice is more closer to the real world than the square lattice. The triangular lattice does not exhibit the parity problem [1, 12].

The *protein folding problem* is to find a folding (conformation) of a protein sequence such that the number of contacts is maximum (or the overall energy is minimum). For a variety of such simple models, this minimization problem is NP-complete [5, 8, 19].

For a protein sequence S, let N_h denote the number of H's in S, N_p denote the number of P's in S, and N_{ph} be the number of (PH)'s in S. If $N_p \geq N_h$ and $N_{ph} > \frac{N_h}{2}$, then we call S a *sparse sequence*; otherwise, it is called a *dense sequence.*

In [1], Agarwala *et al.* proposed a set of folding rules on a triangular lattice and analyzed their approximation ratios. In particular, on 2D triangular lattice, they proposed some folding rules, namely, *backbone folding* which has $\frac{1}{4}$-approximation, *improved backbone folding* which has $\frac{1}{2}$-approximation, *arrow folding* which has $\frac{1}{2}$-approximation, and *improved arrow folding* which has $\frac{6}{11}$-approximation. Unfortunately, the worse cases of these foldings occur in sparse sequences. Besides, by exploring PDB (Protein Data Bank), we find that there are 10788 (among 24615) sequences which are sparse. It motivates this study.

In this paper, we propose two folding rules, namely, *starlike folding* and *compact starlike folding* for sparse sequences. The starlike folding yields $\frac{7}{12}$-approximation and the compact starlike folding yields better than $\frac{7}{12}$-approximation. The rest of this paper is organized as follows: Section 2 provides a necessary background. Section 3 presents our main results. Finally, we give some concluding remarks in Section 4.

2 Preliminaries

A graph of triangular lattice is a placement of vertices on distinct points of the regular triangular lattice such that each edge of the graph maps to adjacent node on the lattice. Every node in the lattice has 6 neighbors.

A protein sequence in HP model is represented as $S = s_1s_2s_3s_4\cdots s_n$, where $s_i \in \{\mathrm{H,P}\}$ for all $1 \leq i \leq n$. A folding of a protein in the 2D HP model is an embedding of S in the 2D lattice. Two H-nodes in lattice form a *non-local H-H bond* if they are adjacent in the lattice but not adjacent in S. This non-local H-H bond is called a *contact* [2]. That is, if s_i and s_j are adjacent H-nodes in lattice and $|i-j| > 1$, then there is a contact between s_i and s_j. It is not hard to see that an H-node has at most 4 contacts except its neighbors in the sequence. If an H-node has 4 contacts, this node is called a *complete node.*

For each of the neighboring lattice points that is occupied by neither an adjacent node in the sequence nor an H-node, we call this lattice edge a *missing contact.* Therefore, for each node of the sequence on triangular lattice, the sum of the contacts and missing contacts is at most 4, and is 5 if this node is the endpoint of the sequence.

2.1 Backbone Folding

Backbone folding is a simple folding strategy based on laying out two backbones anti-parallel to each other such that most of the contacts are achieved among the H's along a backbone and the H's across the backbones [1]. It is according to the following rules to fold a protein sequence.

- Every H is placed on the backbone.
- Two P's on the backbone never form contacts.

Each node of the backbone must have two neighbors from the other backbone and at least one of which must be an H-node. All the H-nodes are on the backbone and each one must create at least one bond. It gives a minimum of $\frac{N_h}{2}$ bonds out of a maximum of $2N_h$ bonds. Therefore, the approximation ratio of the backbone folding is at least $\frac{1}{4}$ [1].

For some special sequences, the backbone folding has a near optimal solution. For example, consider the sequence $S = (\mathrm{HP})^n$, we can use the backbone folding to layout this sequence. In this folding, most of the H-nodes have 4 contacts. The four nodes at the endpoints of the backbone strip have only 2 or 3 contacts. Thus, the number of contacts is $C(S) = 2N_h - 3$. It almost yields an optimal folding in the 2D HP model on the triangular lattice for large values of n.

2.2 Arrow Folding

The following definitions come from [1]. A *strip* is a maximal substring of a sequence containing only H's. For example, HHHH is a 4-strip starting at position 2 of $S =$ PHHHHPHPPPPPHH-PHHHHH. A *separator* is a maximal substring of a sequence containing only P's and a *k-separator* likewise. For example, PPPP is a 4-separator between 1-strip and 2-strip in S.

Define levels L^{+1}, L^0 and L^{-1}, a sequence is folded along this pathway such that all the H-nodes are distributed only in L^{+1}, L^0, and L^{-1}. All the residues on level L^0 are H-nodes. Arrow folding yields a $\frac{1}{2}$-approximation in the 2D HP model on triangular lattice for large values of N_h [1].

In an arrow folding, the arrow can be divided into three groups, head, body, and tail. It is not hard to see that the arrow tail is a sparse sequence of H's and most of the missing contacts are in the arrow tail. In this area, all H's are placed in L^0 level and each H has only 2 contacts according to the rule of arrow folding. In summary, the arrow folding is suitable for dense sequences.

3 Main Results

In this section, we propose some local folding rules for the 2D HP model on triangular lattices. Together with backbone and arrow folding rules, we propose a general strategy for using these rules to fold a protein sequence.

3.1 Circle Folding and Hexagon Folding

At first, we introduce a local rule for folding H^n with a large n. If an H-node is on the border of a folding graph of H^n, then this H-node is a non-complete node. All of the complete nodes must be in the internal part of a folding graph. Therefore, if we want to increase the number of contacts, we have to reduce the number of H-nodes on the border of the folding graph.

If we apply arrow folding on H^n, then its shape looks like a bar. It is known that the circumference of a circle is smaller than the perimeter of any other shape with the same area [2]. That is, if H^n is folded as a circle, it can get the maximum number of contacts. Thus, the *circle folding* is a folding such that the resulting folding graph is a circle.

Though circle folding is better than arrow folding in long H^n, it is hard to implement. We now propose another folding similar to the circle folding. We call it *hexagon folding.* The hexagon folding is easier to construct than circle folding because the circle folding has to calculate the length of each chord. Hexagon folding only surrounds the central node and change its direction on different level from center to outside. Furthermore, its folding graph is equivalent to circle folding in short sequence. So we use hexagon folding instead of circle folding.

3.2 Starlike Folding

In this section, we focus on the sparse sequences. Consider the case that a lot of (HP^+)'s or (H^2P^+)'s are in a sequence. If we use the backbone folding or the arrow folding, then we obtain a result near $\frac{1}{2}$-approximation. Suppose that we apply the arrow folding to a sparse sequence of H's like $(HP^+)^+$, it is not hard to see that all the H's are placed on a line

Now we introduce *starlike folding* for sparse sequences. It is named because its folding shape looks like "$*$." The idea is to put H's together and put P's outside H's. It is similar to backbone folding, with a slight modification for sequences of (HP^+)'s. The difference between starlike folding and backbone folding is the rule for folding P's. It defines six directions like a star, one for pathway in and out, the other directions for folding the P's. A sequence is folded along this pathway such that all the H's are placed only in central area.

Note that for each direction of a star determines a ray for a sequence of P's. In the case of long P chains, the rays from multiple stars will intersect. It can be handled by operations similar to *pull moves* proposed by Lesh *et al.* [13].

Now we analyze the approximation ratio of starlike folding for sparse sequences. Let S be such a sequence. Divide S into k subsequences $S_1, S_2, \ldots, S_k$, i.e., $S = S_1 S_2 \cdots S_k$. Each S_i contains 6 H's except S_k which may contain less than 6 H's. Let $N_{h(i)}$ (respectively, $N_{ph(i)}$) be the number of H's (respectively, (PH)'s) in S_i. Let $N_{d(i)} = N_{h(i)} - N_{ph(i)}$. For each S_i, $1 \leq i \leq k$, the number of contacts is $C(S_i) = 2N_{h(i)} - 3 - N_{d(i)}$. By the definition, $N_{ph(i)} > \frac{N_h(i)}{2}$. Thus, $N_{d(i)} \leq$

$\frac{N_{h(i)}}{2} - 1$. It leads to $C(S_i) = 2N_{h(i)} - 3 - N_{d(i)} \geq 2N_{h(i)} - 3 - \frac{N_{h(i)}}{2} + 1 = \frac{3N_{h(i)}}{2} - 2$. Since $k \leq \frac{N_h}{6} + 1$, $C(S) = \sum_{i=1}^{k}(\frac{3N_{h(i)}}{2} - 2) = \frac{3N_h}{2} - 2k \geq \frac{3N_h}{2} - 2(\frac{N_h}{6} + 1) = \frac{7N_h}{6} - 2$. Therefore, the approximation ratio is $\frac{C(S)}{2N_h} \geq \frac{\frac{7N_h}{6} - 2}{2N_h} \approx \frac{7}{12}$. Thus, we have the following theorem.

Theorem 3.1 *Starlike folding yields a $\frac{7}{12}$-approximation for sparse sequences.*

It shows that the starlike folding is better than arrow folding on sparse sequences. Note that the starlike folding is not suitable for dense sequences. An $H^n (n > 1)$ subsequence in a dense sequence may decrease the approximation. In this case, there exists an $H^n, n > 2$, between two starlike foldings. This H^n is too short to be folded, so it may decrease the approximation. Thus, arrow folding is a good choice. In the following section we propose the compact starlike folding to tackle this problem.

3.3 The Compact Starlike Folding

Although the starlike folding increases the approximation for sparse sequences, a short H^n ($3 \leq n$) between two starlike folding may decrease the approximation. For tackling this problem, we use starlike folding to surround this short H^n in order to increase the folding approximation. We call this rule *compact starlike folding*.

Now we describe the rules of the compact starlike folding. In the case of $3 \leq n \leq 7$, then we place the short H^n in a hexagon. We select one node on the hexagon as a start point and select one neighboring node as end point in order to place more HP^+ surrounding the short H^n. This little adjustment is different from the rule of hexagon folding in the first level of hexagon.

Second, we place the other H's (with the form of HP^+) outside the H^n. The P's are placed outside these H's according to the rules of starlike folding. Every H in HP^+ must be kept close to the inner part of starlike folding.

Consider an example to compare the effect of these folding rules. Let $S = PHP^2HP^2H^2P^2HP^3H^{19}P^2HP^2H^2P^2HP^2HP^2$. Table 1 shows the result.

The approximation of the compact starlike folding is hard to measure. However, in our experiment, its approximation is better than others for sparse sequences. Hence, we conjecture that its approximation is better than others for sparse sequences.

Folding rules	*Contacts*	*App.*
Arrow folding	32	0.55
starlike folding	36	0.62
Compact starlike folding	47	0.81

Table 1: A result of different folding rules.

Sequence types	*Folding rules*
$N_h > N_p$	Arrow
$N_h = N_p$, $N_t = 1$	Arrow
$N_h = N_p = N_t$	Backbone
$N_h < N_p$, $N_t \doteqdot 1$	Arrow
$N_h < N_p$, $N_t \doteqdot N_h$	Starlike
$N_h < N_p$, $N_t \doteqdot N_h$, $N_{H^3} > 0$	Compact starlike

Table 2: N_h: no. of H nodes, N_p: no. of P nodes, N_t: no. of (PH)'s, N_{H^3}: no. of (H^3)'s.

3.4 A General Strategy

Finally, we integrated these rules mentioned above and propose a general strategy for folding a protein sequence on triangular lattice in the 2D HP model. Our strategy is as follows.

Step 1. Find all the subsequences H^n with $n \geq 8$ and fold them by hexagon folding or arrow folding.

Step 2. For each remaining subsequence, choose a folding rule to fold it according to the suggestion listed in Table 2.

Step 3. Check each pair of adjacent foldings and try to use the compact starlike folding to combine them.

Step 4. Combine each folded sequence sequentially.

4 Conclusion

In this paper, we propose the starlike folding and the compact starlike folding on triangular lattice in the 2D HP model. For sparse sequences, starlike folding yields $\frac{7}{12}$-approximation and compact starlike folding seems better than starlike folding. Both of them are better than the previous folding rules for sparse sequences. Together with backbone folding and arrow folding, we suggest a general strategy for folding any protein sequence on triangular lattice in the 2D HP model. For the future, we will consider the same problem on 3D triangular lattice and consider feasible global rules.

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